

Solution Requirements

Date	29 June 2026
Team ID	XXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Solution Requirements

The system accepts agricultural input data such as soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH level, and rainfall. After validating the input data, the machine learning model analyzes the environmental conditions using historical crop dataset and predicts the most suitable crop for cultivation. The predicted result is then displayed through a simple web interface for farmers or users.

Functional Requirements

Requirement ID	Requirement	Description
FR-01	User Input	The system shall allow users to enter parameters for crop prediction.
FR-02	Data Validation	The system shall validate the entered agricultural data before processing.
FR-03	Dataset Access	The system shall retrieve historical crop dataset for training and analysis.
FR-04	Prediction	The machine learning model shall predict the best crop based on input conditions.
FR-05	Display Result	The system displays the recommended crop to the user through the interface.

Non-Functional Requirements

Requirement ID	Requirement	Description
NFR-01	Performance	The system should generate crop predictions within a few seconds.
NFR-02	Accuracy	The model should provide high accuracy in crop recommendation results.
NFR-03	Reliability	The system should consistently produce stable and correct predictions.
NFR-04	Security	User input data should be processed securely without data loss.
NFR-05	Usability	The application should provide a simple and user-friendly interface.

SoftwareRequirements

Software	Purpose
Python	Programming language
Jupyter Notebook	Model development and testing
VSCode / PyCharm	Code development
Flask	Web application framework
Pandas	Data manipulation
NumPy	Numerical computations
Scikit-learn	Machine learning algorithms

Matplotlib	Data visualization
Seaborn	Statistical visualization

Hardware Requirements

Component	Specification
Processor	Intel i3 or above
RAM	Minimum 4 GB
Storage	Minimum 10 GB Free Space
Operating System	Windows / Linux / macOS
Internet	Required for dataset and packages download